

Laboratorio 8

Problema 1

```
void function(int n) {  
    int i, j, k, counter = 0;  
    for (i = n/2; i <= n; i++) {  
        for (j = 1; j <= n/2; j++) {  
            for (k = 1; k <= n; k = k+2) {  
                counter++;  
            }  
        }  
    }  
}
```

a) Encontrar complejidad de tiempo en notación Big-Oh

Ciclo de i

$$N_i = n - \left\lfloor \frac{n}{2} \right\rfloor + 1 = \left\lceil \frac{n}{2} \right\rceil + 1 \rightarrow \text{No. de iteraciones} = \Theta(n)$$

• Ciclo de j

$$j \leq \left\lfloor \frac{n}{2} \right\rfloor \leq n$$

$$j \leq n - \left\lceil \frac{n}{2} \right\rceil = \left\lfloor \frac{n}{2} \right\rfloor$$

$$N_j = \left\lfloor \frac{n}{2} \right\rfloor \rightarrow \text{No. de iteraciones} = \Theta(n)$$

• Ciclo de k

$$2^r \leq n \Rightarrow r \leq \log_2 n \quad 1, 2, 4, 8, \dots, 2^r$$

$$N_k = (\log_2 n) + 1 \rightarrow \text{No. de iteraciones} = \Theta(\log(n))$$

• Complejidad Total

$$C(n) = \left(\left\lceil \frac{n}{2} \right\rceil + 1 \right) \left\lfloor \frac{n}{2} \right\rfloor (\log_2 n + 1)$$

$$T(n) = \Theta(n) \cdot \Theta(n) \cdot \Theta(\log n) = \Theta(n^2 \log n)$$

$\hookrightarrow O(n^2 \log n)$

Problema 2

```
if (n <= 1) return;
```

```
for (i = 1; i <= n; i++) {
    for (j = 1; j <= n; j++) {
        print f("Sequence\n");
        break;
    }
}
```

(El break limita el ciclo interior a una iteración)

• Condición inicial
 $T(n) = \Theta(1) \quad n \leq 1$

• Complejidad Total
 $T(n) = a + bn$

• Ciclo exterior
 El ciclo de i va de 1 a n, realiza n iteraciones

$T(n) = \Theta(n)$

$T(n) = O(n)$ //

Problema 3

```
for (i = 1; i <= n/3; i++) {
    for (j = 1; j <= n; j += 4)
        print f("Sequence\n");
}
```

• Ciclo exterior
 $N_i = \left\lfloor \frac{n}{3} \right\rfloor \rightarrow \text{No. iteraciones} = \Theta(n)$

• Ciclo interior

j : 1, 5, 9, 13

Para $n \geq 1$: $N_j = \left\lfloor \frac{n-1}{4} \right\rfloor + 1 = \left\lfloor \frac{n}{4} \right\rfloor \rightarrow \text{No. iteraciones} = \Theta(n)$

• Complejidad Total

$C(n) = \left\lfloor \frac{n}{3} \right\rfloor \left\lfloor \frac{n}{4} \right\rfloor \rightarrow \frac{n}{3} \cdot \frac{n}{4} = \frac{n^2}{12}$

$T(n) = \Theta(n^2)$

$T(n) = O(n^2)$ //

Problema 4 - Mejor caso, caso promedio y peor caso

A. Linear Search

Mejor caso

$$C_{\text{mejor}}(n) = 1$$

$$T_{\text{mejor}}(n) = \Theta(1)$$

Peor caso

$$C_{\text{peor}}(n) = n$$

$$T_{\text{peor}}(n) = \Theta(n)$$

Caso Promedio

$$E[C(n)] = \frac{1}{n} \sum_{i=1}^n i$$

$$\sum_{i=1}^n i = \frac{n(n+1)}{2} \rightarrow E[C(n)] = \frac{1}{n} \cdot \frac{n(n+1)}{2} = \frac{n+1}{2}$$

$$T_{\text{promedio}}(n) = \Theta(n)$$

B. Binary Search

Mejor Caso

$$T_{\text{mejor}}(n) = \Theta(1)$$

Peor Caso

$$\frac{n}{2^k}$$

$$\frac{n}{2^k} \leq 1 \Rightarrow k \geq \log_2 n \rightarrow T_{\text{peor}}(n) = \Theta(\log n)$$

Caso promedio

$$E[C(n)] = \frac{\sum_{d=0}^h (d+1) 2^d}{n} \rightarrow \sum_{d=0}^h (d+1) 2^d = h 2^{h+1} + 1$$

$$E[C(n)] = \frac{h 2^{h+1} + 1}{2^{h+1} - 1} = \Theta(h) \rightarrow T_{\text{promedio}}(n) = \Theta(\log n)$$

C. Quicksort

Mejor Caso

$$T(n) = 2T(n/2) + \Theta(n)$$

$$T_{\text{mejor}}(n) = \Theta(n \log n)$$

Peor Caso

$$T(n) = T(n-1) + \Theta(n)$$

$$n + (n-1) + (n-2) + \dots + 1 = \frac{n(n+1)}{2}$$

$$T_{\text{peor}}(n) = \Theta(n^2)$$

Caso promedio

$$E[T(n)] = \frac{1}{n} \sum_{k=1}^{n-1} (E[T(k)] + E[T(n-1-k)]) + \Theta(n)$$

$$E[T(n)] = \frac{2}{n} \sum_{k=1}^{n-1} E[T(k)] + \Theta(n)$$

$$E[C(n)] = (n+1) H_n - n$$

$$H_n = \sum_{i=1}^n \frac{1}{i} = \Theta(\log n)$$

$$T_{\text{promedio}}(n) = \Theta(n \log n)$$

Problema 5 - Verdadero o falso

a) Si $f(n) = \Theta(g(n))$ y $g(n) = \Theta(h(n))$, entonces $h(n) = \Theta(f(n))$

R// Verdadero

$$f(n) = \Theta(g(n))$$

$$ag(n) \leq f(n) \leq bg(n)$$

Para todo $n \geq n_1$

$$g(n) = \Theta(h(n))$$

$$ch(n) \leq g(n) \leq dh(n)$$

$$h(n) = \Theta(f(n))$$

Para $n \geq \max(n_1, n_2)$

$$ach(n) \leq f(n) \leq bdh(n)$$

$$\frac{1}{bd} f(n) \leq h(n) \leq \frac{1}{ac} f(n)$$

b) Si $f(n) = O(g(n))$ y $g(n) = O(h(n))$, entonces $h(n) = \Omega(f(n))$

R// Verdadero

$0 \leq f(n) \leq ag(n)$ → Por definición de Big-Oh, existen constantes positivas a, b

$0 \leq g(n) \leq bh(n)$

$f(n) \leq abh(n)$

$h(n) \geq \frac{1}{ab} f(n)$ → Cumple la definición de cota inferior asintótica

$$h(n) = \Omega(f(n))$$

c) $f(n) = \Theta(n^2)$ donde $f(n)$ está definido como el tiempo de ejecución

def A(n):
 tuples = tuple(range(0, n))
 S = set()
 for i in range(0, n):
 for j in range(i+1, n):
 S.add(tuple([i, j]))

R// Falso

$$\Theta(n)$$

$$\Theta(1)$$

$$\sum_{i=0}^{n-1} (n-i-1) = \frac{n(n-1)}{2} = \Theta(n^2)$$

Costo de tupla [i:j]

$L = j - i$

$\Theta(j - i)$

Suma del trabajo

$$\sum_{i=0}^{n-2} \sum_{j=i+1}^{n-1} (j-i)$$

$$\sum_{i=0}^{n-1} (j-i) = \frac{(n-1-i)(n-i)}{2}$$

$$\sum_{m=1}^{n-1} \frac{m(m+1)}{2} = \frac{1}{2} \left(\sum_{m=1}^{n-1} m^2 + \sum_{m=1}^{n-1} m \right)$$

$$= \frac{1}{2} \left(\frac{(n-1)n(2n-1)}{6} + \frac{(n-1)n}{2} \right)$$

$$= \frac{n(n-1)(n+1)}{6} = \frac{n^3 - n}{6}$$

$$T(n) = \Theta(n^3)$$